

FYUGP Internship Assessment Test – Data Science

1.MCQ Assessment Link:

<https://forms.gle/Qv7bL5Xn6s9mkumz9>

2.Python

Write a Python program to:

1. Take two numbers from the user
2. Print their sum, subtraction, multiplication, and division

3.Python Lists

Create a list containing 5 student names and:

1. Print the list
2. Add one new name
3. Remove one name

4.SQL

Create a table named **students** with the following columns:

- id
- name
- age
- course

5.SQL Query

Write an SQL query to:

1. Display all records from the **students** table
2. Display students whose age is greater than 20

6.NumPy

Write a NumPy program

1. that creates a 2D array x of shape (3, 4) and a 1D array y of shape (4,).
2. Subtract y from each row of x using broadcasting.

7.Pandas

Create a dataframe with the following columns: name, age, and gender. The dataframe should have 10 rows of data.

1. Add a new column to the data frame created in question 1, called occupation. The values for this column should be Programmer, Manager, and Analyst, corresponding to the rows in the dataframe.
2. Select the rows of the dataframe where the age is greater than or equal to 30.
3. Convert this dataframe to a CSV file and read that CSV file, and finally display the contents.

8. Matplotlib

Create a line chart using Matplotlib for:

- X values → [1,2,3,4,5]
- Y values → [10,20,30,40,50]

Add:

- Title
- X-axis label
- Y-axis label

Assessment Rules

1. MCQ Assessment Score: 25 Marks
2. Questions 2 to 4 carry 3 marks each.
3. Questions 5 to 8 carry 4 marks each.
4. Total Marks: 50

